

INTEGRATED AGRI EXPORT & COLD CHAIN CONTROL TOWER · MUMBAI / JNPA

AMAMA TRADECHAIN

Nine operating views, captured from the live React build

01 Executive — board & investors

02 Field & procurement — farmers, QC, lots

03 Mumbai hub — warehouse & cold chain

04 Freight control tower — vessels & cut-offs

05 Documentation — files & customs

06 Buyer portal — orders & traceability

07 Exceptions — faults & evidence

08 Responsibility matrix — 28 controlled documents

09 Integration layer — 17 adapters

- Executive

Board & Investors
- Field & procurement

Farmers, QC, lots
- Mumbai hub

Warehouse & cold chain
- Freight control tower

Vessels & cut-offs
- Documentation

Files & customs
- Buyer portal

Orders & traceability
- Exceptions

Faults & evidence
- Responsibility matrix

28 controlled documents
- Integration layer

17 adapters

THROUGHPUT THIS SEASON

6,410 MT

Across 11 commodity lines

EXPORT CONTAINERS SHIPPED

257 TEU

9 destination countries

ON-TIME SAILING RATE

96 %

Lane benchmark 78%

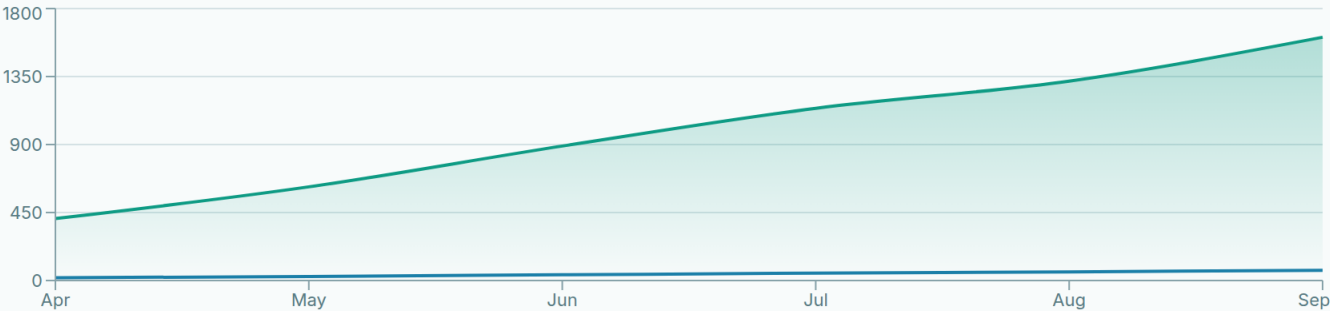
POST-HARVEST LOSS

6.1 %

Baseline 18–22% for the same lanes

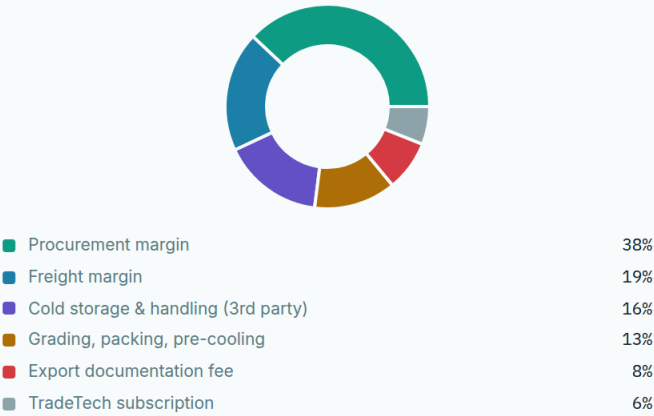
THROUGHPUT & CONTAINER VELOCITY

Tonnes handled and containers sailed per month



REVENUE MIX

Ten streams, one asset base

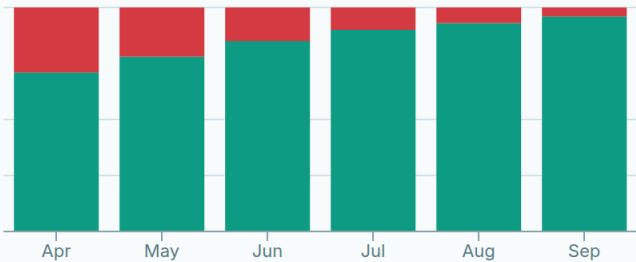


EVENT FEED

Control tower online · 6 active export files

SAILINGS KEPT VS ROLLED

The metric the whole platform exists to move



LIVE SHIPMENT BOARD

6 active files · 1 critical

EXP-DXB-POM-2026-0087	Pomegranate — Bhagwa · 20 MT · Jebel Ali, UAE	MSC Aurora	T-41h	21	ON TRACK
EXP-AMS-GRP-2026-0091	Grapes — Thompson Seedless · 16 MT · Rotterdam, NL	Maersk Cape Coral	T-27h	86	CRITICAL
EXP-SIN-BAN-2026-0093	Banana — Grand Naine · 24 MT · Singapore	CMA CGM Dutch Mate	T-76h	21	ON TRACK
EXP-DOH-MNG-2026-0095	Mango — Alphonso · 9 MT · Hamad Port, QA	Maersk Cape Coral	T-27h	16	ON TRACK
EXP-JED-POM-2026-0098	Pomegranate — Bhagwa · 22 MT · Jeddah, KSA	ONE Harbour	T-128h	25	ON TRACK
EXP-LGP-ONI-2026-0101	Onion — Nashik Red · 27 MT · Colombo, LK	COSCO Fortune	T-151h	24	ON TRACK

FARMERS ONBOARDED

5,140

22 FPOs across 6 districts

AVERAGE FARMGATE UPLIFT

+17%

Versus local mandi rate, same grade, same day

OPEN EXCEPTIONS

0

Median time to close 4.2 h

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PRE-HARVEST FORECAST

What the system knows 30–60 days before the farmer calls

FARMER	DISTRICT	CROP & VARIETY	EXPECTED	WINDOW	QUALITY HISTORY	FPO
Sanjay Patil · F-001	Nashik	Pomegranate — Bhagwa	25 MT	12–18 Sep	94%	—
Rekha Deshmukh · F-002	Sangli	Grapes — Thompson	18 MT	09–14 Sep	91%	Krishna Valley FPO
Imran Shaikh · F-003	Jalgaon	Banana — Grand Naine	40 MT	15–25 Sep	88%	Tapi Growers FPO
Vaibhav More · F-004	Solapur	Pomegranate — Bhagwa	31 MT	18–24 Sep	79%	—
Anita Kale · F-005	Nashik	Onion — Nashik Red	55 MT	05–20 Sep	85%	Godavari FPO
Prakash Yadav · F-006	Ratnagiri	Mango — Alphonso	12 MT	01–08 Sep	96%	Konkan Alphonso FPO

Forecast rows drive truck rostering, cold room allocation and vessel booking before harvest. A trader reacts to supply; this reacts to a forecast.

LOTS CREATED TODAY

Every consignment carries its own identity from this point forward

No lots yet. Grade a sample on the left to open the first lot of the day.

FIELD QC & LOT CREATION

Collection centre — Dindori, Nashik

FARMER

Sanjay Patil — Pomegranate — Bhagwa

QUANTITY OFFERED — 12 MT

SAMPLE INSPECTION

Size uniformity

Colour & maturity

Brix

Physical damage

Bruising / rot

Pest damage

Foreign matter

Pulp temperature

Packaging condition

Attach sample photos

Auto-assigned grade

CREATE LOT & PRINT LABEL

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INBOUND DOCK CONTROL

No truck arrives unannounced — every vehicle holds a slot

MH-15-EK-8842

Sanjay Patil · AMAMA-POM-MH-2026-000127

ETA 12 min 5.4°C

In transit

ASSIGN DOCK

MH-04-CJ-2019

Rekha Deshmukh · AMAMA-GRP-MH-2026-000144

ETA 48 min 1.9°C

In transit

ASSIGN DOCK

MH-19-BQ-7731

Anita Kale (FPO) · AMAMA-ONI-MH-2026-000180

ETA arrived

At gate

ASSIGN DOCK

MH-12-AZ-5560

Prakash Yadav · AMAMA-MNG-MH-2026-000162

ETA 95 min 12.6°C

Loading

ASSIGN DOCK

HUB ZONE UTILISATION

Zone A → L · receiving through dispatch

A

Truck receiving

5/8 slots

B

QC laboratory

3/6 slots

C

Sorting & grading

9/12 slots

D

Washing & processing

2/6 slots

E

Pre-cooling

8/10 pallets

F

Cold storage

664/900 pallets

G

Dry storage

173/400 pallets

H

Packaging

11/14 slots

I

Palletisation

6/10 slots

J

Export inspection

3/4 slots

K

Container stuffing

4/6 slots

L

Dispatch

2/6 slots

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DXB-POM-2026-0087

21 · T-41h

AMS-GRP-2026-0091

86 · T-27h

SIN-BAN-2026-0093

21 · T-76h

DOH-MNG-2026-0095

16 · T-27h

JED-POM-2026-0098

25 · T-128h

LGP-ONI-2026-0101

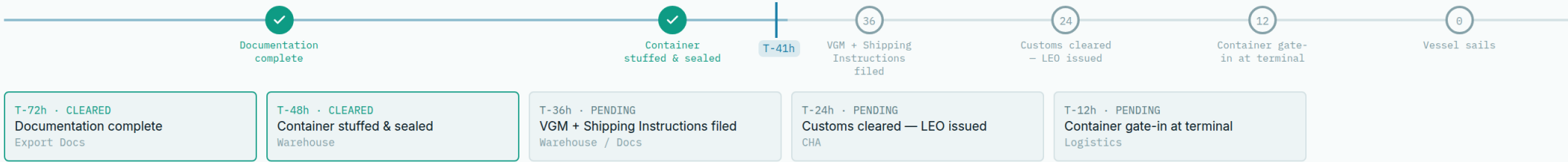
24 · T-151h

EXP-DXB-POM-2026-0087

Pomegranate — Bhagwa · 20 MT · 2 × 40'RH @ 5°C · Al Maha Fresh FZC, Jebel Ali, UAE

21

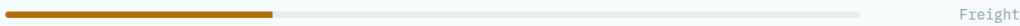
ON TRACK



WHY THIS SHIPMENT IS AT RISK

Weighted drivers, recomputed every second

MSC rollover history 24% on this trade lane



VGM + Shipping Instructions filed — only 5 h of slack



REHEARSE A DISRUPTION

VESSEL OMISSION

REEFER EXCURSION

CUSTOMS QUERY

CUT-OFF ADVANCED

EXCEPTION ENGINE

RUN RISK REVIEW

VESSEL LADDER

Primary, backup and emergency tonnage held open on every lane — switching is one click, not one week

6 sailings in window

	CARRIER / VESSEL	VOYAGE	ETD	TRANSIT	ROUTING	REEFER PLUGS	USD/40RH	ROLLOVER HISTORY	FIT	
BACKUP 1	CMA CGM Dutch Mate	0AB2X	+82 h	8 d	Direct	60	\$2340	9%	79	SWITCH
BACKUP 2	Hapag-Lloyd Bremen Express	114W	+106 h	9 d	Jebel Ali T/S	31	\$2410	7%	67	SWITCH
STANDBY	COSCO Fortune	091W	+154 h	10 d	Direct	55	\$1890	19%	57	SWITCH
STANDBY	ONE Harbour	022E	+130 h	11 d	Colombo T/S	24	\$2050	15%	44	SWITCH
STANDBY	Maersk Cape Coral	0426W	+34 h	6 d	Direct	42	\$2180	11%	35	SWITCH
BOOKED	MSC Aurora	2609E	+58 h	7 d	Direct	18	\$1960	24%	25	

Fit blends rollover history, declared reefer plugs, routing, transit days and slack against the T-72 rule. Cargo is never held hostage to a single booking — the moment a primary sailing degrades, the backup is already priced, plugged and quoted.

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SHIPMENT DOCUMENT FILE

Nothing gates in until this file closes — the checklist is the control, not the reminder

EXP-DXB-POM-2026-0087

43%

12/28

STATUS	STAGE	DOCUMENT	RESPONSIBLE (R)	ACCOUNTABLE (A)	CONSULTED (C)	SYSTEM OF RECORD	DUE
CLEARED	Procurement	Purchase order / procurement confirmation	Procurement Officer	Head — Procurement	Finance	Procurement module	Pre-harvest
CLEARED	Procurement	Farmer invoice & weighment slip	Collection Centre In-charge	Head — Procurement	Finance	Farmer app	At pickup
CLEARED	Farm QC	Field quality inspection report	Field QC Inspector	Head — Quality	Procurement	QC module	At pickup
CLEARED	Farm QC	Grading sheet & Lot ID allotment	Field QC Inspector	Head — Quality	—	Traceability	At pickup
CLEARED	Farm QC	Farm registration / traceability record (APEDA HortiNet, where applicable)	Compliance Executive	Head — Compliance	Field QC	HortiNet API	Pre-season
CLEARED	Transport	E-way bill & transport document	Logistics Coordinator	Head — Logistics	Finance	TMS / NIC API	Before dispatch
CLEARED	Transport	Loading confirmation + seal record + photos	Collection Centre In-charge	Head — Logistics	QC	TMS	At loading
CLEARED	Hub intake	Gate-in slip & dock QC report	Warehouse Supervisor	Hub Manager	QC	WMS	On arrival
CLEARED	Export QC	Laboratory / residue test report	Compliance Executive	Head — Quality	Empanelled lab	LIMS integration	T-96 h
CLEARED	Export QC	Pre-shipment inspection certificate	Export QC Officer	Head — Quality	Inspection agency	QC module	T-72 h
CLEARED	Commercial	Sales contract / proforma invoice	Export Sales Executive	Head — Sales	Finance	Order module	T-21 d
CLEARED	Commercial	Commercial invoice	Export Documentation Exec	Manager — Docs	Finance	Docs module	T-72 h
OPEN	Commercial	Packing list	Export Documentation Exec	Manager — Docs	Warehouse	WMS → Docs	T-72 h
OPEN	Origin	Certificate of Origin	Export Documentation Exec	Manager — Docs	Chamber / DGFT portal	eCoO portal	T-72 h
OPEN	Plant health	Phytosanitary certificate (where applicable)	Compliance Executive	Head — Compliance	Plant Quarantine	PQIS	T-72 h
OPEN	Plant health	Fumigation certificate (where applicable)	Compliance Executive	Head — Compliance	Accredited operator	Docs module	T-72 h
OPEN	Freight	Carrier booking confirmation	Freight Executive	Manager — Freight	Sales	Carrier eBooking API	T-14 d
NOT STARTED	Freight	Container PTI report (reefer)	Freight Executive	Manager — Freight	Carrier depot	Depot feed	T-60 h
NOT STARTED	Freight	Shipping instructions (SI)	Export Documentation Exec	Manager — Docs	Freight	Carrier API / INTTRA	T-36 h
NOT STARTED	Terminal	VGM declaration	Warehouse Supervisor	Hub Manager	Freight	Weighbridge → carrier	T-36 h
NOT STARTED	Customs	Shipping bill filing	CHA / Customs Broker	Manager — Docs	Compliance	ICEGATE	T-48 h
NOT STARTED	Customs	e-SANCHIT supporting document upload	Export Documentation Exec	Manager — Docs	CHA	ICEGATE e-SANCHIT	T-48 h
NOT STARTED	Customs	Let Export Order (LEO)	CHA / Customs Broker	Manager — Docs	—	ICEGATE	T-24 h
NOT STARTED	Terminal	Gate-in / terminal acceptance record	Logistics Coordinator	Head — Logistics	Terminal	Terminal feed	T-12 h
NOT STARTED	Post-shipment	Bill of Lading	Freight Executive	Manager — Freight	Docs	Carrier API	T+48 h
NOT STARTED	Post-shipment	Marine insurance certificate	Finance Executive	Head — Finance	Insurer	Finance module	T-24 h
NOT STARTED	Post-shipment	LC / bank negotiation set	Finance Executive	Head — Finance	Bank	Trade finance	T+72 h
NOT STARTED	Post-shipment	Buyer traceability pack (QR dossier)	Traceability Analyst	Head — Technology	QC	Traceability	At sailing

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ORDER BOOK

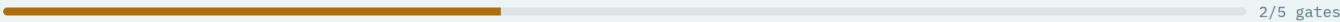
What the buyer sees, without a phone call

EXP-DXB-POM-2026-0087

Watch

ETA 9 days

Pomegranate — Bhagwa · 20 MT · Grade A+ · AI Maha Fresh FZC



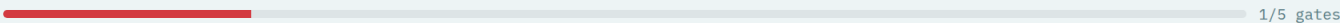
Lots: AMAMA-POM-MH-2026-000127 · AMAMA-POM-MH-2026-000131

EXP-AMS-GRP-2026-0091

Critical

ETA 7 days

Grapes — Thompson Seedless · 16 MT · Grade A+ · Vanderveen Produce BV



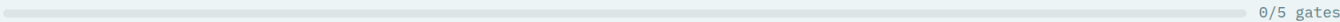
Lots: AMAMA-GRP-MH-2026-000144

EXP-SIN-BAN-2026-0093

On track

ETA 11 days

Banana — Grand Naine · 24 MT · Grade A · Sembawang Fruits Pte



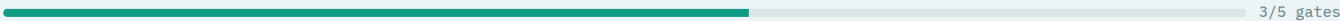
Lots: AMAMA-BAN-MH-2026-000158

EXP-DOH-MNG-2026-0095

On track

ETA 7 days

Mango — Alphonso · 9 MT · Grade A+ · Qatar Gourmet Trading



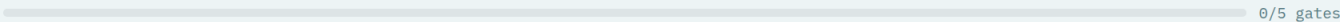
Lots: AMAMA-MNG-MH-2026-000162

EXP-JED-POM-2026-0098

On track

ETA 16 days

Pomegranate — Bhagwa · 22 MT · Grade A · Reef Al Sharq Co.



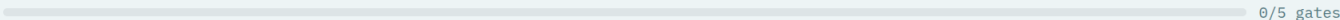
Lots: AMAMA-POM-MH-2026-000171

EXP-LGP-ONI-2026-0101

On track

ETA 16 days

Onion — Nashik Red · 27 MT · Grade B · Ceylon Agri Imports



Lots: AMAMA-ONI-MH-2026-000180

SCAN A LOT

Every carton carries a QR that opens this record

AMAMA-POM-MH-2026-000127



- FARMER**
Sanjay Patil · F-001 · Dindori, Nashik, Maharashtra
- HARVEST**
10 September 2026 · hand-picked, morning window
- FIELD QC**
Grade A+ · 9/9 parameters cleared · 4 sample photos on file
- COLLECTION**
Dindori centre · weighbridge slip #WB-40218 · e-way bill raised
- TRANSIT**
Reefer MH-15-EK-8842 · 5.2–5.6°C held · door sealed, 0 openings
- HUB INTAKE**
Mumbai hub · dock QC cleared · pre-cooled to core 5°C in 6 h 40 m
- STORAGE**
CR-01 · 5°C / 92% RH · 148 pallets · 31 h dwell
- EXPORT QC**
Residue panel cleared · pre-shipment inspection certificate issued
- CONTAINER**
2 × 40'RH · PTI passed · set point 5°C · logger armed
- VESSEL**
MSC Aurora 2609E · ex JNPA
- DESTINATION**
Jebel Ali, UAE · AI Maha Fresh FZC

- 

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Documentation

Files & customs
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Buyer portal

Orders & traceability



Exceptions

Faults & evidence



Responsibility matrix

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Control tower online · 6 active export files

REPORT A FAULT

Any stakeholder, any point on the chain, with photo evidence

LOCATION

Cold storage — Zone F

FAULT TYPE

Reefer unit alarm

SEVERITY

High

WHAT DID YOU SEE

Describe the fault and anything already done about it.

 Attach photo evidence

RAISE FAULT

EXCEPTION QUEUE

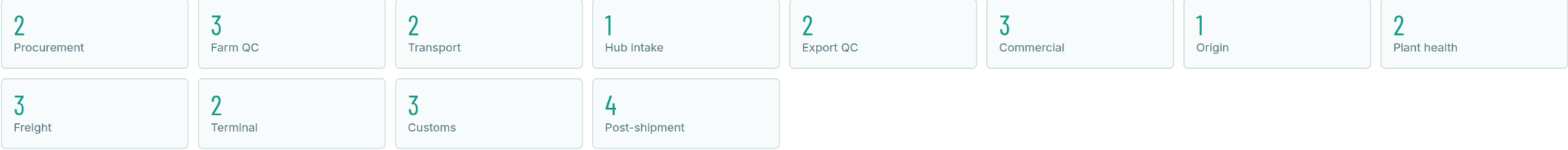
Routed by location to a named owner, with a clock running

Queue is clear. Raise a fault on the left to see it route to an owner.

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DOCUMENTATION & RESPONSIBILITY MATRIX								
28 controlled documents · R = does the work, A = answerable for it, C = consulted, I = informed								
Q Filter								
#	STAGE	DOCUMENT	R — RESPONSIBLE	A — ACCOUNTABLE	C — CONSULTED	I — INFORMED	SYSTEM	DUE
01	Procurement	Purchase order / procurement confirmation	Procurement Officer	Head — Procurement	Finance	Farmer / FPO	Procurement module	Pre-harvest
02	Procurement	Farmer invoice & weighment slip	Collection Centre In-charge	Head — Procurement	Finance	Farmer	Farmer app	At pickup
03	Farm QC	Field quality inspection report	Field QC Inspector	Head — Quality	Procurement	Warehouse	QC module	At pickup
04	Farm QC	Grading sheet & Lot ID allotment	Field QC Inspector	Head — Quality	—	Sales	Traceability	At pickup
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08	Hub intake	Gate-in slip & dock QC report	Warehouse Supervisor	Hub Manager	QC	Procurement	WMS	On arrival
09	Export QC	Laboratory / residue test report	Compliance Executive	Head — Quality	Empanelled lab	Buyer	LIMS integration	T-96 h
10	Export QC	Pre-shipment inspection certificate	Export QC Officer	Head — Quality	Inspection agency	Buyer	QC module	T-72 h
11	Commercial	Sales contract / proforma invoice	Export Sales Executive	Head — Sales	Finance	Buyer	Order module	T-21 d
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13	Commercial	Packing list	Export Documentation Exec	Manager — Docs	Warehouse	CHA	WMS → Docs	T-72 h
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17	Freight	Carrier booking confirmation	Freight Executive	Manager — Freight	Sales	Warehouse	Carrier eBooking API	T-14 d
18	Freight	Container PTI report (reefer)	Freight Executive	Manager — Freight	Carrier depot	Warehouse	Depot feed	T-60 h
19	Freight	Shipping instructions (SI)	Export Documentation Exec	Manager — Docs	Freight	Carrier	Carrier API / INTTRA	T-36 h
20	Terminal	VGM declaration	Warehouse Supervisor	Hub Manager	Freight	Carrier, Terminal	Weighbridge → carrier	T-36 h
21	Customs	Shipping bill filing	CHA / Customs Broker	Manager — Docs	Compliance	Finance	ICEGATE	T-48 h
22	Customs	e-SANCHIT supporting document upload	Export Documentation Exec	Manager — Docs	CHA	Customs	ICEGATE e-SANCHIT	T-48 h
23	Customs	Let Export Order (LEO)	CHA / Customs Broker	Manager — Docs	—	Control tower	ICEGATE	T-24 h
24	Terminal	Gate-in / terminal acceptance record	Logistics Coordinator	Head — Logistics	Terminal	Freight	Terminal feed	T-12 h
25	Post-shipment	Bill of Lading	Freight Executive	Manager — Freight	Docs	Buyer, Bank	Carrier API	T+48 h
26	Post-shipment	Marine insurance certificate	Finance Executive	Head — Finance	Insurer	Buyer	Finance module	T-24 h
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INTEGRATION LAYER

Seventeen adapters behind one internal event bus · the intelligence adapter is live in this build

FREIGHT

Carrier eBooking

Maersk / CMA CGM / Hapag REST

POST /bookings

Creates booking, returns booking no. + cut-offs

ADAPTER READY · 141 ms

Vessel schedules

DCSA OVS 3.0 / INTTRA

GET /schedules

Fills the vessel ladder every 6 h

ADAPTER READY · 244 ms

Track & trace events

DCSA T&T 2.0 webhook

webhook

Gate-in, loaded, departed, discharged

ADAPTER READY · 63 ms

COLD CHAIN

Reefer telemetry

Remote container monitoring

GET /containers/{id}/telemetry

Set point vs supply/return air, 15-min interval

ADAPTER READY · 240 ms

Truck GPS + temp logger

Telematics provider

webhook

Geofence, door open, excursion alerts

ADAPTER READY · 222 ms

Cold room IoT

MQTT broker

subscribe

Per-room temp, RH, door, power

ADAPTER READY · 168 ms

COMPLIANCE

APEDA HortiNet

Farm & lot registry

GET/POST

Farm registration, lot creation, lab linkage

ADAPTER READY · 91 ms

Plant Quarantine PQIS

Phyto application

POST

Phytosanitary certificate workflow

ADAPTER READY · 175 ms

CUSTOMS

ICEGATE e-SANCHIT

Document upload + IRN

POST

Supporting docs, digitally signed

ADAPTER READY · 252 ms

Shipping bill status

ICEGATE enquiry

GET

Assessment, examination, LEO status

ADAPTER READY · 216 ms

FINANCE

GST e-Invoice (IRP)

NIC IRP

POST

IRN + signed QR on export invoice

ADAPTER READY · 161 ms

E-way bill

NIC EWB

POST

Auto-generated on dispatch

ADAPTER READY · 297 ms

Farmer payout

Bank / UPI payout

POST

Same-day settlement against weighment

ADAPTER READY · 92 ms

Trade finance / LC

Bank corporate API

GET

LC status, document discrepancies

ADAPTER READY · 179 ms

MARKET

Mandi price feed

Agmarknet

GET

Domestic fallback pricing for rejected grades

ADAPTER READY · 293 ms

Farmer notifications

WhatsApp Business API

POST

Pickup slot, QC result, payment advice

ADAPTER READY · 218 ms

INTELLIGENCE

Risk & exception engine

Anthropic Claude API

POST /v1/messages

Live in this build — reads shipment state, returns mitigation plan

LIVE · 70 ms

EVENT CONTRACT

One canonical event shape, so any new carrier or terminal is an adapter and not a rebuild

```
POST /v1/events
{
  "event_id":   "evt_01J9K4M2XR",
  "type":       "container.gate_in",           // 42 typed events across 6 domains
  "occurred_at": "2026-09-18T04:12:33+05:30",
  "source":     "terminal.jnpa",               // adapter that emitted it
  "refs": {
    "shipment": "EXP-DXB-POM-2026-0087",
    "lots":     ["AMAMA-POM-MH-2026-000127"],
    "container": "MSKU7841203",
    "booking":  "MAEU248811907"
  },
  "payload": { "gate": "P4", "vgm_kg": 24180, "seal": "IN0448212" },
  "risk_delta": -14                               // fed straight into the score
}
```

Every adapter normalises into this shape before anything downstream sees it. The risk score, the buyer portal, the farmer's WhatsApp message and the investor dashboard all read the same stream — which is why the numbers on every screen agree.